Week 2: Paper Writing Practice

MCM Preparation

Abstract

This practice document connects a simple question, an assumption, a model, and its results. We ask how far an object travels at a constant speed of 2 metres per second. Using a constant-speed model, we obtain distances of 10, 20, and 30 metres after 5, 10, and 15 seconds. A table and a figure present the same calculated values. The model applies only when the speed remains constant. All values are illustrative; this is a typesetting exercise rather than a report of measured data.

1 Introduction

The reading exercise for this week uses a 2023 MCM paper by Team 2314151 [1]. The example in this document uses a separate, elementary motion problem to practise document structure. Our question is: how far does an object travel in a specified time when its speed is constant?

2 Assumptions

1. The object moves along a straight path in one direction.
2. Its speed remains constant during the interval considered.
3. Distance and elapsed time are both zero at the starting point.

These assumptions let us use a single speed parameter. Acceleration and stops are outside the scope of this example.

3 Notation

Symbol	Meaning	Unit
t	Elapsed time	s
v	Constant speed	m/s
$s(t)$	Distance travelled by time t	m

4 Model and Solution

Under the stated assumptions, distance equals speed multiplied by time:

$$s(t) = vt, \qquad t \geq 0. \tag{1}$$

Here $v = 2$ m/s is an illustrative input. Substituting $t = 10$ s into Equation (1) gives $s(10) = 20$ m. Each result therefore has a stated input, a calculation, and a unit.

5 Results

Table 1: Calculated distances for the illustrative constant-speed model.

Time t (s)	Distance $s(t)$ (m)
5	10
10	20
15	30

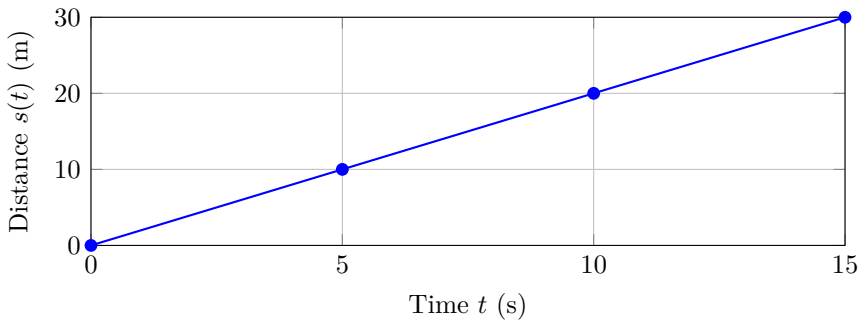

Figure 1: Distance increases linearly with time in the illustrative model.

Table 1 lists three calculated values, and Figure 1 shows their common linear trend. At 10 seconds, the calculated distance is 20 metres. These values follow from the assumptions; they do not validate the model against observations.

6 Sensitivity and Model Evaluation

For a fixed time $t > 0$, Equation (1) implies that a 10% increase in speed produces a 10% increase in calculated distance. This describes the response to one parameter under the same assumptions.

The model is easy to interpret, and its units are consistent. Its main limitation is that changes in speed are ignored. A real application would need observations to assess whether the constant-speed assumption is appropriate and to measure prediction error.

7 Conclusion

With an illustrative speed of 2 m/s, the model gives a distance of 20 m after 10 s. The result answers the original question only under the constant-speed assumption. In a modelling paper, the abstract, model, tables, figures, and conclusion should tell this same story.

References

[1] MCM Team 2314151. Breaking the Wordle. 2023 Mathematical Contest in Modeling, Problem C, 2023. URL https://reformship.github.io/pages/3competition/4mcm/MCM%20Outstanding/2023/C/2314151.pdf. Reading sample; full text hosted by a third-party mirror.